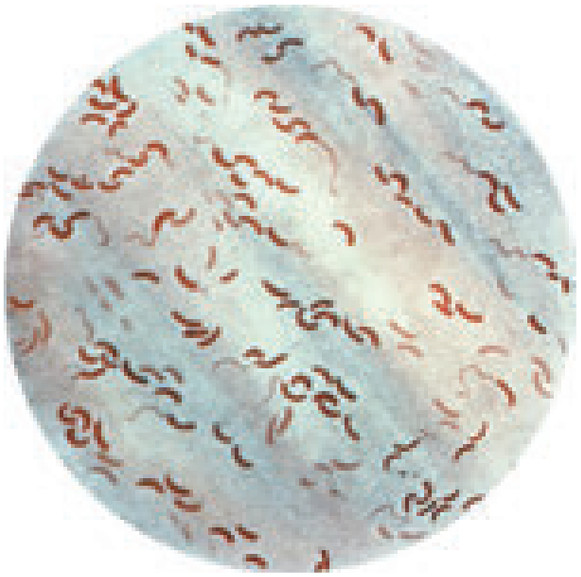


Figure 7.16: Gram staining of *Vibrio cholerae*



Figure 7.17: Colony morphology of *Vibrio cholerae* on TCBS

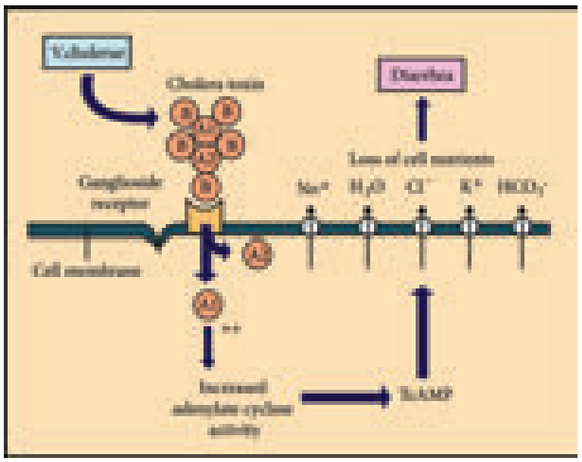


Figure 7.18: Mechanism of action of Cholera toxin

7.10.3 Enterotoxin

Vibrios multiplying on the intestinal epithelium produce an enterotoxin called **Cholera toxin**. It is also known as **Choleragen** (or CT). This toxin molecule is approximately 84,000 Dalton and consists of two major subunits namely A and B There is only one subunit in A (1A) whereas there are five subunits in B (5B) (Figure 7.18).

Mode of Action

- The B (binding) units of enterotoxin get attached to the GM_1 (Ganglioside membrane receptors I) on the surface of jejunal epithelial cells (target cells).
- The A (active) subunits then enters the target cell and dissociates into 2 fragments, A_1 & A_2. The A_2 fragment links biologically active A_1 fragment to the B – subunit.
- The A_1 fragment causes prolonged activation of cellular adenylate cyclase which in turn accumulates cAMP in the target cell. This leads to outpouring of large quantities of water and electrolytes into small intestinal lumen. Thus, resulting in profuse watery diarrhea.



Natural infection of *Vibrio cholerae* occurs only in human beings

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